

Agenda

1. Announcements

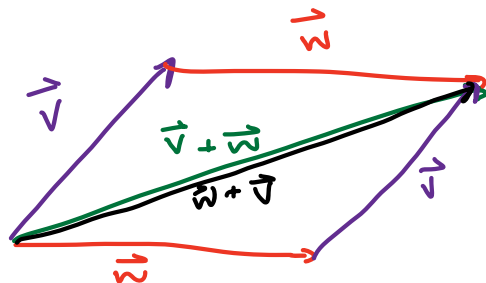
- Regrade Requests due next Wednesday
- Drop deadline is today

2.) Finish 12.2

3.) Start 12.3

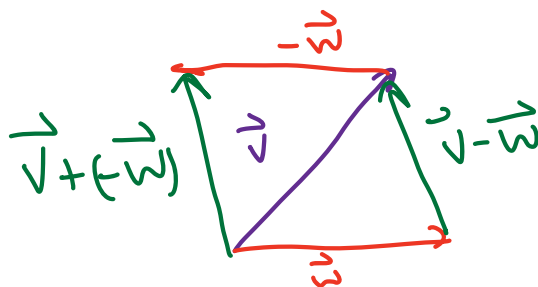
Parallelogram Law

For any vectors $\vec{v}$ and $\vec{w}$, we get $\vec{v} + \vec{w} = \vec{w} + \vec{v}$



The difference of two vectors $\vec{v}$ and $\vec{w}$ is defined as

$$\vec{v} - \vec{w} = \vec{v} + (-\vec{w})$$



Vector Length

Given $\vec{v} = \langle x, y, z \rangle$, its length or magnitude is given as

$$|\vec{v}| = \sqrt{x^2 + y^2 + z^2} = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2}$$

$\|\vec{v}\|$ the dist. between $(0,0,0)$ and (x,y,z)

Activity: Find the indicated vector lengths:

1.) $|\langle 1, 2, 3 \rangle|$

$$l = \sqrt{1^2 + 2^2 + 3^2}$$

$$= \sqrt{1 + 4 + 9} = \sqrt{14}$$

2.) $|\langle \frac{3}{4}, -\frac{1}{4} \rangle|$

$$l = \sqrt{(\frac{3}{4})^2 + (-\frac{1}{4})^2}$$

$$= \sqrt{\frac{9}{16} + \frac{1}{16}} = \frac{\sqrt{10}}{4}$$

we call $\vec{u}$ a unit vector when its magnitude is 1, $|\vec{u}| = 1$
Standard basis for $\mathbb{R}^3$: $\vec{i} = \langle 1, 0, 0 \rangle$, $\vec{j} = \langle 0, 1, 0 \rangle$, $\vec{k} = \langle 0, 0, 1 \rangle$

Properties of Vectors

Let $\vec{v}$ and $\vec{w}$ be vectors. Let c and d be constants.
 $\text{in } \mathbb{R}^2 \text{ or } \mathbb{R}^3$ Scalars

- | | |
|--|--|
| 1.) $\vec{v} + \vec{w} = \vec{w} + \vec{v}$ | 2.) $\vec{v} + \vec{0} = \vec{v}$ |
| 3.) $\vec{v} - \vec{w} = \vec{v} + (-\vec{w}) = \vec{0}$ | 4.) $c(\vec{v} + \vec{w}) = c\vec{v} + c\vec{w}$ |
| 5.) $(c+d)\vec{v} = c\vec{v} + d\vec{v}$ | 6.) $1\vec{v} = \vec{v}$ |
| 7.) $0\vec{v} = \vec{0}$ | |

Example: Let $\vec{v} = \langle -1, 5, 6 \rangle$ and $\vec{w} = \langle 3, 0, 4 \rangle$. Find:

- 1.) $5\vec{v} - 3\vec{w} + 4\vec{v} = (5 - 3 + 4)\vec{v} = 6\vec{v} = 6\langle -1, 5, 6 \rangle = \langle -6, 30, 36 \rangle$
- 2.) $0(\vec{v} + \vec{w}) = \langle 0, 0, 0 \rangle = \vec{0}$

Dot Product Let $\vec{v} = \langle x_1, y_1, z_1 \rangle$ and $\vec{w} = \langle x_2, y_2, z_2 \rangle$

be vectors in $\mathbb{R}^3$. The dot product is:

$$\vec{v} \cdot \vec{w} = \langle x_1, y_1, z_1 \rangle \cdot \langle x_2, y_2, z_2 \rangle$$

$$\vec{v} \cdot \vec{w} = x_1 x_2 + y_1 y_2 + z_1 z_2$$

Example:

$$1.) \langle 1, 2, 3 \rangle \cdot \langle -1, 1, -1 \rangle = -1 + 2 + -3 = -2$$

$$2.) \langle 1, 3 \rangle \cdot \langle -5, 6 \rangle = -5 + 18 = 13$$

Activity: $\langle 2, 3, -2 \rangle \cdot \langle 6, -2, 3 \rangle$

$$12 + -6 + -6 = 0$$

Properties of the dot product

Let $\vec{v}$, $\vec{w}$, and $\vec{u}$ be vectors. Let c be a constant scalar

$$1.) \vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$$

$$2.) \vec{v} \cdot (\vec{w} + \vec{u}) = \vec{v} \cdot \vec{w} + \vec{v} \cdot \vec{u}$$

$$3.) c(\vec{v} \cdot \vec{w}) = c\vec{v} \cdot \vec{w} = c\vec{w} \cdot \vec{v}$$

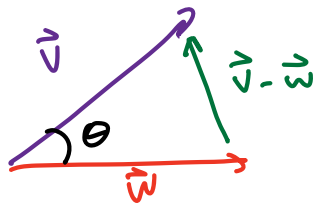
$$4.) \vec{0} \cdot \vec{v} = 0$$

$$5.) \vec{v} \cdot \vec{v} = |\vec{v}|^2$$

Proof of 5.)

Angles Between Vectors

Let $\vec{v}$ and $\vec{w}$ be vectors. We define the angle θ , where $0 \leq \theta \leq \pi$ to be the angle between the vectors with the same initial point:



Law of Cosines

$$|\vec{v} - \vec{w}|^2 = |\vec{v}|^2 + |\vec{w}|^2 - 2|\vec{v}||\vec{w}|\cos\theta$$

Theorem: If θ is the angle between vectors $\vec{v}$ and $\vec{w}$, then $\vec{v} \cdot \vec{w} = |\vec{v}||\vec{w}|\cos\theta$

$$\cos\theta = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}||\vec{w}|}$$

Example: Find the angle between $\overbrace{3\vec{i} + 3\vec{j} + 3\vec{k}}^{<3, 3, 3>}$ and $\underbrace{-2\vec{i} - 2\vec{k}}_{<-2, 0, -2>}$

$$\cos\theta = \frac{<3, 3, 3> \cdot <-2, 0, -2>}{\sqrt{3^2 + 3^2 + 3^2} \cdot \sqrt{(-2)^2 + 0^2 + (-2)^2}} = \frac{-6 + 0 - 6}{\sqrt{27} \cdot \sqrt{8}}$$

$$= \frac{-12}{\sqrt{27} \cdot \sqrt{8}}$$

$$\theta = \arccos\left(\frac{-12}{\sqrt{27} \cdot \sqrt{8}}\right)$$

$$\theta = 2.526 \text{ radians}$$

Activity: Find the angle between the vectors

- 1.) $\langle 3, -1 \rangle$ and $\langle -1, 4 \rangle$ 2.) $\langle -1, 3, 2 \rangle$ and $\langle 4, 2, -1 \rangle$

Definition: Two vectors $\vec{v}$ and $\vec{w}$ are orthogonal when the angle between them is $\pi/2$

Theorem: